Compact object headers

One 8-byte header word (JEP 519, JDK 25)

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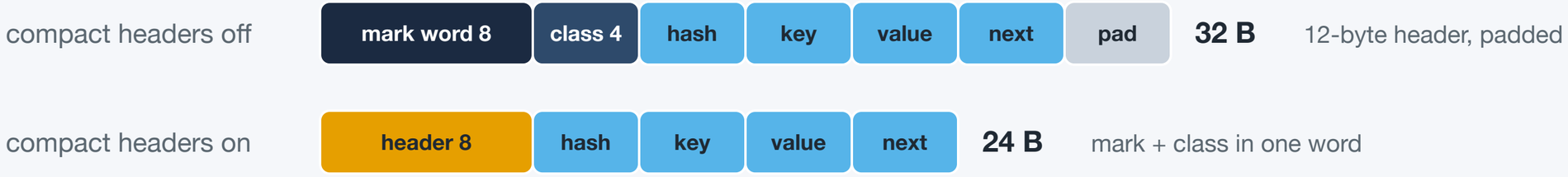
What is a compact object header?

The object header shrinks from 12 bytes to one 8-byte word.

- A header holds the mark word (hash, lock, GC state) and a class pointer.
- Product feature in JDK 25 (JEP 519); experimental in JDK 24 (JEP 450).
- Off by default: `-XX:+UseCompactObjectHeaders`.

Where the 4 bytes matter

HashMap$Node layout (box width = bytes)



1M HashMap entries: 56.4 MB → 48.4 MB. An Integer stays 16 B either way (12 B rounds up to 16).

- An object only shrinks if it then fits a smaller 8-byte boundary.
- Less heap and less GC work for object-heavy data.

References

- JDK 25 Release Notes: JEP 519 Compact Object Headers
- The Arrival of Java 25 (Oracle Java blog)
- The `java` command (Java SE 25)